

Postmastectomy Radiotherapy: An American Society of Clinical Oncology, American Society for Radiation Oncology, and Society of Surgical Oncology Focused Guideline Update Summary

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Abstract

The American Society of Clinical Oncology guideline for the use of postmastectomy radiotherapy (PMRT) was published in 2001. This update of that guideline, completed in collaboration with the American Society for Radiation Oncology and the Society of Surgical Oncology, focuses on key areas of ongoing controversy, including the use of PMRT for patients with one to three positive lymph nodes, use of PMRT for patients undergoing neoadjuvant systemic therapy, and selected technical aspects of PMRT, particularly the extent of regional nodal irradiation. The question of whether PMRT is indicated in women with T1-2 tumors and a positive sentinel node biopsy who do not undergo completion axillary lymph node dissection is also discussed.

The “Postmastectomy Radiotherapy: An American Society of Clinical Oncology, American Society for Radiation Oncology, and Society of Surgical Oncology Focused Guideline Update,” an update of the 2001 American Society of Clinical Oncology guideline,¹ was developed and written by Abram Recht, Elizabeth A. Comen, Richard E. Fine, Gini F. Fleming, Patricia H. Hardenbergh, Alice Y. Ho, Clifford A. Hudis, E. Shelley Hwang, Jeffrey J. Kirshner, Monica Morrow, Kilian E. Salerno, George W. Sledge Jr, Lawrence J. Solin, Patricia A. Spears, Timothy J. Whelan, Mark R. Somerfield, and Stephen B. Edge. 

Authors' Disclosures of Potential Conflicts of Interest

Disclosures provided by the authors are available with this article at jop.ascopubs.org.

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DOI: [10.1200/JOP.2016.016469](https://doi.org/10.1200/JOP.2016.016469);
published online ahead of print at
jop.ascopubs.org on September 20,
2016.

THE BOTTOM LINE

Postmastectomy Radiotherapy: An American Society of Clinical Oncology, American Society for Radiation Oncology, and Society of Surgical Oncology Focused Guideline Update

Guideline Questions and Target Populations

- Is postmastectomy radiotherapy (PMRT) indicated in patients with T1-2 tumors with one to three positive axillary lymph nodes who undergo axillary lymph node dissection (ALND)?
- Is PMRT indicated in patients with T1-2 tumors and a positive sentinel node biopsy (SNB) who do not undergo completion ALND?
- Is PMRT indicated in patients presenting with clinical stage I or II cancers who have received neoadjuvant systemic therapy (NAST)?
- Should regional nodal irradiation (RNI) include the internal mammary (IMNs) and/or supraclavicular-axillary apical nodes when PMRT is used in patients with T1-2 tumors with one to three positive axillary nodes?

Target Audience

Medical oncologists, radiation oncologists, surgeons, pathologists, oncology nurses, patients, and caregivers

Clinical Question 1

Is PMRT indicated in patients with T1-2 tumors with one to three positive axillary lymph nodes who undergo ALND?

Recommendations

Recommendation 1a.

The panel unanimously agreed that the available evidence shows that PMRT reduces the risks of locoregional failure (LRF), any recurrence, and breast cancer mortality for patients with T1-2 breast cancer and one to three positive lymph nodes. However, some subsets of these patients are likely to have such a low risk of LRF that the absolute benefit of PMRT is outweighed by its potential toxicities. In addition, the acceptable ratio of benefit to toxicity varies among patients and physicians. Thus, the decision to recommend PMRT or not requires a great deal of clinical judgment. The panel agreed clinicians making such recommendations for individual patients should consider factors that may decrease the risk of LRF, attenuate the benefit of reduced breast cancer–specific mortality, and/or increase the risk of complications resulting from PMRT. These factors include: patient characteristics (age > 40 to 45 years, limited life expectancy because of older age or comorbidities, or coexisting conditions that might increase the risk of complications), pathologic findings associated with a lower tumor burden (eg, T1 tumor size, absence of lymphovascular invasion, presence of only a single positive node and/or small size of nodal metastases, or substantial response to NAST), and biologic characteristics of the cancer associated with better outcomes and survival and/or greater effectiveness of systemic therapy (eg, low tumor grade or strong hormonal sensitivity; type: informal consensus; evidence quality: intermediate; strength of recommendation: moderate). There are several risk-adaptive models that physicians may find useful in explaining the benefits of PMRT during shared decision making with patients. However, the panel found insufficient evidence to endorse any specific model or to unambiguously define specific patient subgroups to which PMRT should not be administered (type: no recommendation; evidence quality: low; strength of recommendation: weak). Further research is needed on how to accurately estimate individuals' risk of LRF and hence their potential reductions in LRF and breast cancer mortality.

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Recommendation 1b.

The decision to use PMRT should be made in a multidisciplinary fashion through discussion among providers from all treating disciplines early in a patient's treatment course (soon after surgery or before or soon after the initiation of systemic therapy), either in the context of a formal tumor board or by referral (type: informal consensus; evidence quality: insufficient; strength of recommendation: strong).

Recommendation 1c.

Decision making must fully involve the patient, whose values as to what constitutes sufficient benefit and how to weigh the risk of complications against this in light of the best information the treating physicians can provide regarding PMRT in her situation must be respected and incorporated into the final treatment choice (type: informal consensus; evidence quality: insufficient; strength of recommendation: strong).

Clinical Question 2

Is PMRT indicated in patients with T1-2 tumors and a positive SNB who do not undergo completion ALND?

Recommendation

For patients with clinical T1-2 tumors with clinically negative nodes, SNB is now generally performed at the time of mastectomy, with omission of ALND if the nodes are negative. ALND has generally been performed if the nodes are positive, but there is increasing controversy about whether this is always necessary, especially if there is limited disease in the affected nodes. The panel recognizes that some clinicians omit axillary dissection with one or two positive sentinel nodes in patients treated with mastectomy. This practice is primarily based on extrapolation of data from randomized trials of patients treated exclusively or predominantly with breast-conserving surgery and whole-breast irradiation or breast plus axillary irradiation. In such cases where clinicians and patients elect to omit axillary dissection, the panel recommends that these patients receive PMRT only if there is already sufficient information to justify its use without needing to know that additional axillary nodes are involved (type: informal consensus; evidence quality: weak; strength of recommendation: moderate).

Clinical Question 3

Is PMRT indicated in patients with clinical stage I or II cancers who have received NAST?

Recommendation

Patients with axillary nodal involvement that persists after NAST (eg, less than a complete pathologic response) should receive PMRT. Observational data suggest a low risk of locoregional recurrence for patients who have clinically negative nodes and receive NAST or who have a complete pathologic response in the lymph nodes with NAST. However, there is currently insufficient evidence to recommend whether PMRT should be administered or can be routinely omitted in these groups. The panel recommends entering eligible patients in clinical trials that examine this question (type: informal consensus; evidence quality: low; strength of recommendation: weak).

Clinical Question 4

Should RNI include both the IMNs and supraclavicular-axillary apical nodes when PMRT is used in patients with T1-2 tumors with one to three positive axillary nodes?

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Recommendation

The panel recommends treatment generally be administered to both the IMNs and the supraclavicular-axillary apical nodes in addition to the chest wall or reconstructed breast when PMRT is used for patients with positive axillary lymph nodes. There may be subgroups that will experience limited, if any, benefits from treating both these nodal areas compared with treating only one or perhaps treating only the chest wall or reconstructed breast. There is insufficient evidence at this time to define such subgroups in detail. Additional research is needed to identify them (type: informal consensus; evidence quality: intermediate; strength of recommendation: moderate).

More information, including a Data Supplement, a Methodology Supplement, slide sets, and clinical tools and resources, is available at www.asco.org/pmrt-guideline and www.asco.org/guidelineswiki. Patient information is available at www.cancer.net.

The American Society of Clinical Oncology believes that cancer clinical trials are vital to inform medical decisions and improve cancer care and that all patients should have the opportunity to participate.

References

1. Recht A, Edge SB, Solin LJ, et al: Postmastectomy radiotherapy: Clinical practice guidelines of the American Society of Clinical Oncology. *J Clin Oncol* 19:1539-1569, 2001

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Consulting or Advisory Role: CareCore, US Oncology

Research Funding: Genomic Health (Inst)

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No relationship to disclose

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No relationship to disclose